

Overview of Multi-armed bandit

Subject: Multi-armed bandit

1.

In probability theory and machine learning, the multi-armed bandit problem (sometimes called the K- or N-armed bandit problem) is named from imagining a gambler at a row of slot machines (sometimes known as "one-armed bandits"), who has to decide which machines to play, how many times to play each machine and in which order to play them, and whether to continue with the current machine or try a different machine. More generally, it is a problem in which a decision maker iteratively selects one of multiple fixed choices (i.e., arms or actions) when the properties of each choice are only partially known at the time of allocation, and may become better understood as time passes. A fundamental aspect of bandit problems is that choosing an arm does not affect the properties of the arm or other arms. Instances of the multi-armed bandit problem include the task of iteratively allocating a fixed, limited set of resources between competing (alternative) choices in a way that minimizes the regret. A notable alternative setup for the multi-armed bandit problem includes the "best arm identification (BAI)" problem where the goal is instead to identify the best choice by the end of a finite number of rounds. The multi-armed bandit problem is a classic reinforcement learning problem that exemplifies the exploration–exploitation tradeoff dilemma. In contrast to general reinforcement learning, the selected actions in bandit problems do not affect the reward distribution of the arms. The multi-armed bandit problem also falls into the broad category of stochastic scheduling. In the problem, each machine provides a random reward from a probability distribution specific to that machine, that is not known a priori. The objective of the gambler is to maximize the sum of rewards earned through a sequence of lever pulls. The crucial tradeoff the gambler faces at each trial is between "exploitation" of the machine that has the highest expected payoff and "exploration" to get more information about the expected payoffs of the other machines. The trade-off between exploration and exploitation is also faced in machine learning. In practice, multi-armed bandits have been used to model problems such as managing research projects in a large organization, like a science foundation or a pharmaceutical company. In early versions of the problem, the gambler begins with no initial knowledge about the machines. Herbert Robbins in 1952, realizing the importance of the problem, constructed convergent population selection strategies in "some aspects of the sequential design of experiments". A theorem, the Gittins index, first published by John C. Gittins, gives an optimal policy for maximizing the expected discounted reward.

2.

$\rho = T \mu^* - \sum_{t=1}^T r^t$ $\{\displaystyle \rho = T \mu^* - \sum_{t=1}^T \widehat{r}_t\}$, where μ^* $\{\displaystyle \mu^*\}$ is the maximal reward mean, $\mu^* = \max_k \{\mu_k\}$ $\{\displaystyle \mu^* = \max_k \{\mu_k\}\}$, and r^t $\{\displaystyle \widehat{r}_t\}$ is the reward in round t $\{\displaystyle t\}$. A zero-regret strategy is a strategy whose average regret per round ρ / T $\{\displaystyle \rho / T\}$ tends to zero with probability 1 when the number of played rounds tends to infinity. Intuitively, zero-regret strategies are guaranteed to converge to a (not necessarily unique) optimal strategy if enough rounds are played.

3.

The multi-armed bandit model The multi-armed bandit (short: bandit or MAB) can be seen as a set of real distributions $B = \{R_1, \dots, R_K\}$, each distribution being associated with the rewards delivered by one of the $K \in \mathbb{N}^+$ levers. Let $\mu_1, \dots, \mu_K$ be the mean values associated with these reward distributions. The gambler iteratively plays one lever per round and observes the associated reward. The objective is to maximize the sum of the collected rewards. The horizon H is the number of rounds that remain to be played. The bandit problem is formally equivalent to a one-state Markov decision process. The regret ρ after T rounds is defined as the expected difference between the reward sum associated with an optimal strategy and the sum of the collected rewards:

4.

The multi-armed bandit problem models an agent that simultaneously attempts to acquire new knowledge (called "exploration") and optimize their decisions based on existing knowledge (called "exploitation"). The agent attempts to balance these competing tasks in order to maximize their total value over the period of time considered. There are many practical applications of the bandit model, for example:

5.

Non-stationary bandit This framework refers to the multi-armed bandit problem in a non-stationary setting (i.e., in presence of concept drift). In the non-stationary setting, it is assumed that the expected reward for an arm k can change at every time step $t \in T$: $\mu_{t-1}^k \neq \mu_t^k$. Thus, μ_t^k no longer represents the whole sequence of expected (stationary) rewards for arm k . Instead, μ^k denotes the sequence of expected rewards for arm k , defined as $\mu^k = \{\mu_t^k\}_{t=1}^T$. A dynamic oracle represents the optimal policy to be compared with other policies in the non-stationary setting. The dynamic oracle optimises the expected reward at each step $t \in T$ by always selecting the best arm, with expected reward of μ_t^* . Thus, the cumulative expected reward $D(T)$ for the dynamic oracle at final time step T is defined as:

6.

Infinite-armed bandit In the original specification and in the above variants, the bandit problem is specified with a discrete and finite number of arms, often indicated by the variable K . In the infinite armed case, introduced by Agrawal (1995), the "arms" are a continuous variable in K dimensions.